# Answers
# Exam in Introduction to economics for sport management

**Instructions (read carefully)**

a. **The final exam lasts 3 hours.**
b. **You can use calculators.**
c. **Justify all your answers using economic concepts, theories and reasoning.**

**Good Luck!**

**Question 1 (20 points)**

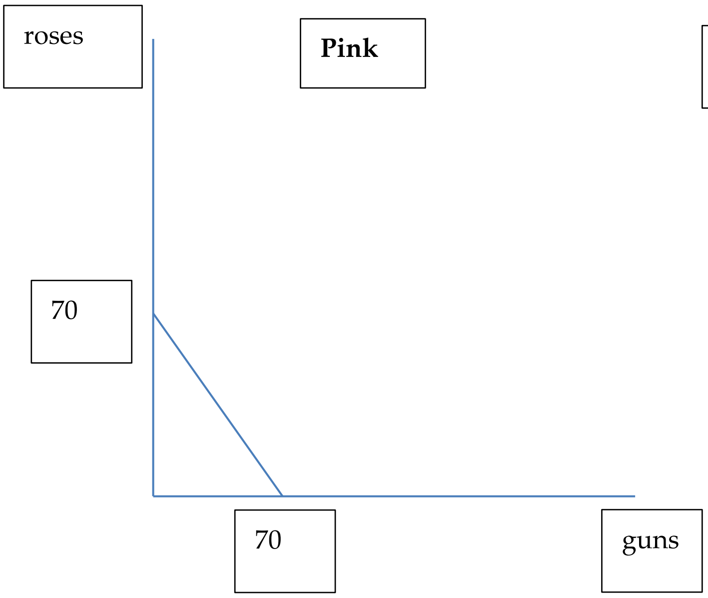
Suppose there are two independent countries Pink and Floyd. Those countries consume two goods, guns (X) and roses (Y).

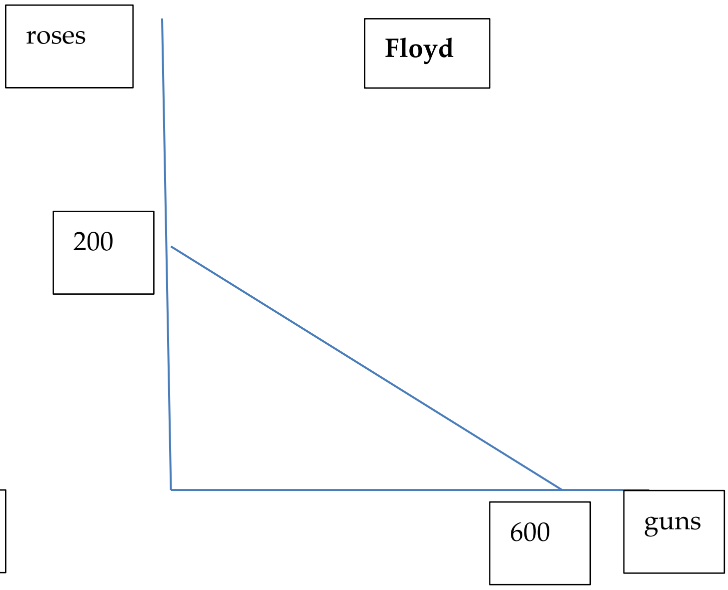
Assume that in Pink there are 70 workers and each of them can produce 1 unit of guns or 1 unit of roses. Assume that in Floyd there are 200 workers and each of them can produce 3 units of guns or 1 unit of roses.

a. Please draw a production possibilities frontier of each of these countries. (6 points)
b. It is known that each country needs to consume 60 guns for self-defence. How many units of roses will be consumed by each country? (4 points)
c. Now assume that there is a world trade, such that each of these countries can sell and buy an infinite number of units of guns and roses as follows: 1 unit of roses is worth 2 units of guns (1 unit of guns is worth 0.5 units of roses). Please draw a production possibilities frontier of each of these countries. (6 points)
d. If each country still needs 60 units of guns, what will be the consumption of roses of each of these countries? (4 points)

**Answers:**

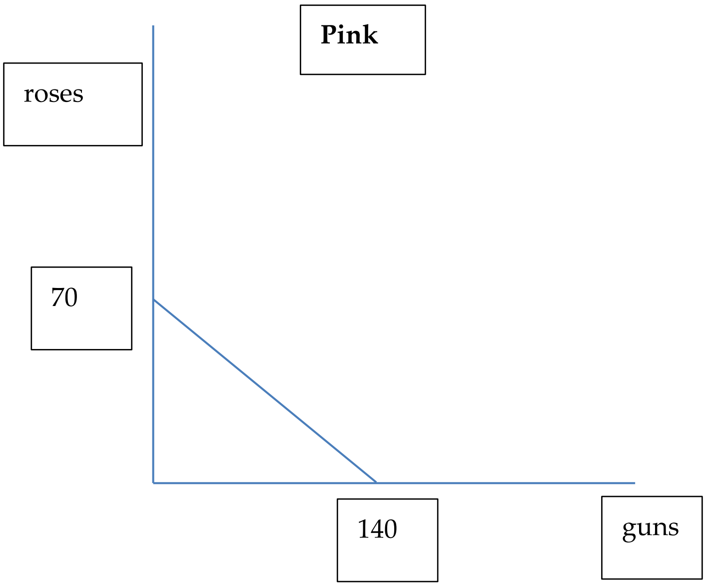
| | Production of guns | Production of roses | MC guns | MC roses |
|---|---|---|---|---|
| Pink | 1 | 1 | 1/1=1 | **1/1=1** |
| Floyd | 3 | 1 | **1/3** | 3 |

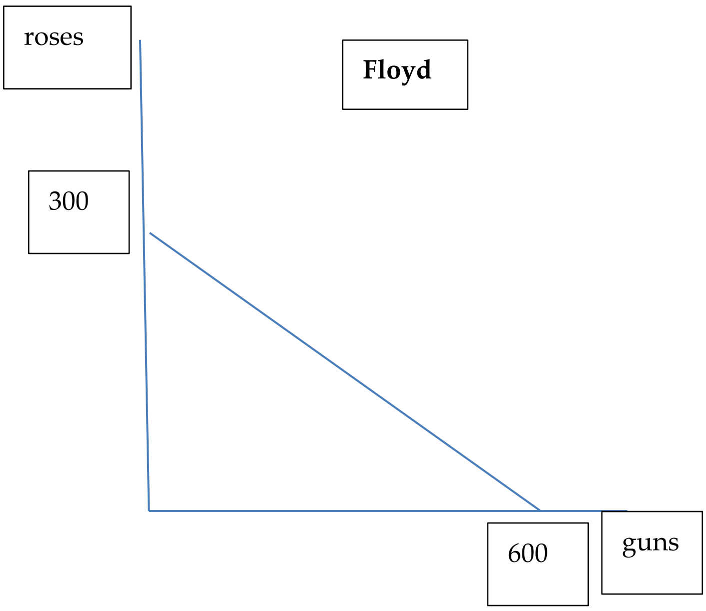



b. To produce 60 units of guns, Pink needs 60 workers. The remaining 10 workers will produce 10 units of butter (10*1). In Floyd, 20 workers will produce 60 guns, and the rest 180 workers (200-20) will produce 180*1=180 units of roses.

c. For Pink, it is worth to produce roses (1 unit of roses costs 1 unit of guns) and sell it to the world for the price of 2 guns per 1 rose. For Floyd, it is worth to produce guns (cost of 1/3 unit of roses per production of 1 unit of gun) and sell it to the world for the price of 0.5 units of roses. Thus, Pink will only produce 70 roses and trade it for guns (at most they can consume 70*2=140 guns). Floyd will only produce 600 guns and trade it for roses (at most they can consume 600*0.5.5=300 roses).



d. If each country still needs 60 units of guns then Floyd will produce 600 units of guns and will sell 540 units of guns for 270 units of roses.

Pink will produce 70 units of roses. They will trade 60 guns for 30 roses and keep to themselves 40 roses (70-30).

## Question 2 (20 points)

Suppose the following demand function: $Q^D$=15-2p. In addition, suppose the following supply function: $Q^S$=p where $Q^D$ is the quantity demanded of a good, $Q^S$ is the supplied quantity of a good and p is the price of a good in USD.

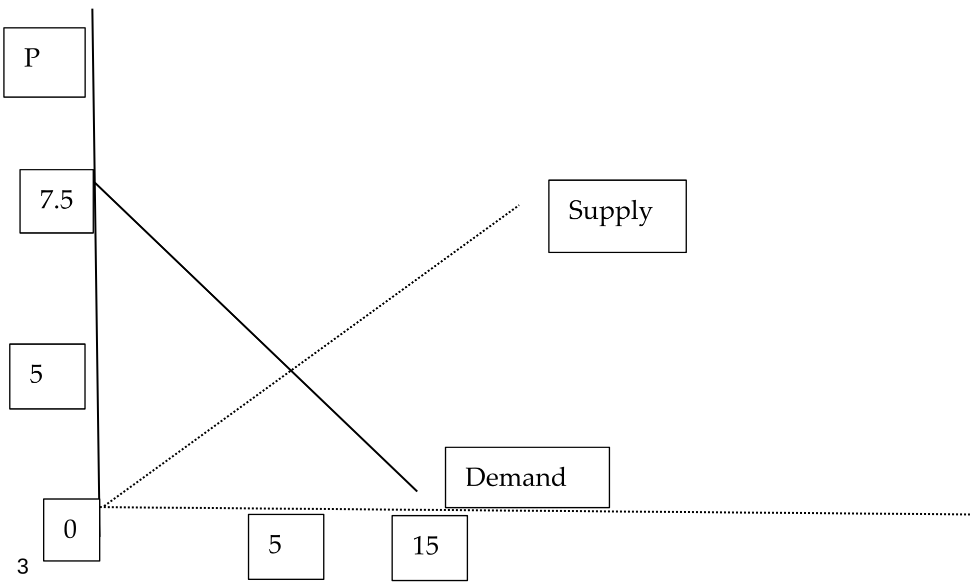
a. What would be the price and the quantity in the equilibrium? **(4 points)**
b. Show on the graph (where appropriate) the cases where p=0, Q=0, and price and quantity in equilibrium. **(8 points)**
c. What is the producer surplus? **(3 points)**
d. What is consumer surplus? **(3 points)**
e. What is the total social welfare? **(2 points)**

**Answer:**

**a. In equilibrium the supply is equal to demand ($Q^S = Q^D$) therefore, 15-2p=p. Solving this equation we get p=5 and Q=5.**

**b. See the answer below**

**c,d,e The total social welfare is the sum of consumer and producer surplus.**
**The consumer surplus is the area between the demand curve and the price (triangle area) (7.5-5)*(5-0)/2=6.25**
**The producer surplus is the area between the supply curve and the price (triangle area) (5-0)*(5-0)/2=12.5**
**The total social welfare is 6.25+12.5=18.75**



## Question 3 (20 points)

It is known that an average firm quits the market in the short term, when the price of a product drops below 15 USD per unit. When the price is above 25 USD per unit, more and more firms enter this market. Based on this information, answer the following questions:
a. What is the minimal AVC of such a firm? **(5 points)**
b. What is the minimal ATC of such a firm? **(5 points)**
c. What is the price that should exist in the long term? **(5 points)**
d. What is the long-term profit of firms if the price is as found in point c? **(5 points)**

**Answers:**

a. **Minimal AVC is 15 USD.**
b. **Minimal ATC is 25 USD**
c. **In the long term, the price should not be above 25 USD**
d. **If p=ATCmin, then the long-term profit is zero.**

## Question 4 (15 points)

There is a market with 8 workers who should be allocated between two fields with the following production function:

Field A

| L | TP |
|---|---|
| 0 | 0 |
| 1 | 5 |
| 2 | 9 |
| 3 | 12 |
| 4 | 14 |
| 5 | 15 |

Field B

| L | TP |
|---|---|
| 0 | 0 |
| 1 | 3 |
| 2 | 7 |
| 3 | 10 |
| 4 | 12 |
| 5 | 13 |

a. How many workers would work on each field if it were known that the price of one produced unit on field A is 2USD and the price of one produced unit on field B is 1USD? **(5 points)**
b. What would be the salary of each worker? **(5 points)**
c. What would be the profit of the owners of each field? **(5 points)**

**Answer:**

Field A

| L | TP | mp | Vmp (p*mp) |
|---|---|---|---|
| 0 | 0 | | |
| 1 | 5 | 5 | 10 |
| 2 | 9 | 4 | 8 |
| 3 | 12 | 3 | 6 |
| 4 | 14 | 2 | 4 |
| 5 | 15 | 1 | 2 |

Field B

| L | TP | mp | Vmp (p*mp) |
|---|---|---|---|
| 0 | 0 | | |
| 1 | 3 | 3 | 3 |
| 2 | 7 | 4 | 4 |
| 3 | 10 | 3 | 3 |
| 4 | 12 | 2 | 2 |
| 5 | 13 | 1 | 1 |

a. **There will be 4 workers in Field A and 4 workers in field A (or 5 workers on field A and 3 workers on field B).**
b. **The value of the marginal product of the last worker is 2 and this will be the wage as well (VMP=W).**
c. **Profit(A with 4 workers)=TP*p-W*L=14*2-2*4=20**
   **Profit(A with 5 workers)=TP*p-W*L=15*2-2*5=20**
   **Profit(B with 4 workers)=TP*p-W*L=12*1-2*4=4**
   **Profit(B with 3 workers)=TP*p-W*L=10*1-2*3=4**

## Question 5 (15 points)

You have the following three saving options:

a. Invest 1000 NOK at the end of each of the next 4 years.
b. Invest 4000 NOK at the end of the second year.
c. Invest 1000 NOK now, 2000 NOK at the end of the second year, 1000 NOK at the end of the fourth year.

What is the best saving option if you wish to withdraw the money at the end of the fifth year and the interest rates are 5% in the first two years, 6% in the third year, 7% in the fourth year, and 8% in the fifth year.

**Answer:**

**FV(a)=1000*1.05*1.06*1.07*1.08+1000*1.06*1.07*1.08+1000*1.07*1.08+1000*1.08= 4746.7**
**FV(b)=4000*1.06*1.07*1.08=4899.7**
**FV(c)= 1000*$(1.05)^2$*1.06*1.07*1.08+2000*1.06*1.07*1.08+1000*1.08= 4880**

**Option B is the best one since it has the highest FV.**

**Question 6 (5 points)**

When would you recommend to tax a certain product? Provide one example of such a good.

**Answer:**

**Taxation is supposed to reduce the quantity consumed. Thus, the recommendation to tax a certain good should be in place if there is a negative externality of a consumption of such a good and we want to reduce its consumption. Sugar related products, cigarettes and alcohol are examples of such products.**

**Question 7 (5 points)**

One of the possible gains from hosting Mega-Events relates to general infrastructure that are politically difficult to provide without the strict deadline of an opening ceremony (roads, airports, etc.). What is the benefit of investing in general infrastructure without hosting mega-events?

**Answer:**

**If a country needs investing in general infrastructure, it can do it without hosting mega-event and without investing in very expensive stadiums. In other words, if you need roads, build roads, why do you need to invest in stadiums for that?**